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CHAPTER 1 • NUMBER • TOPIC 10

Surds

Roots you leave exact. Pull out square factors, expand like algebra, and clear the bottom with the conjugate — DOTS does the work.



$$\frac{3}{10} = 0.3 =$$

30%**5**

VARIANTS

18

PAST-PAPER PARTS

27

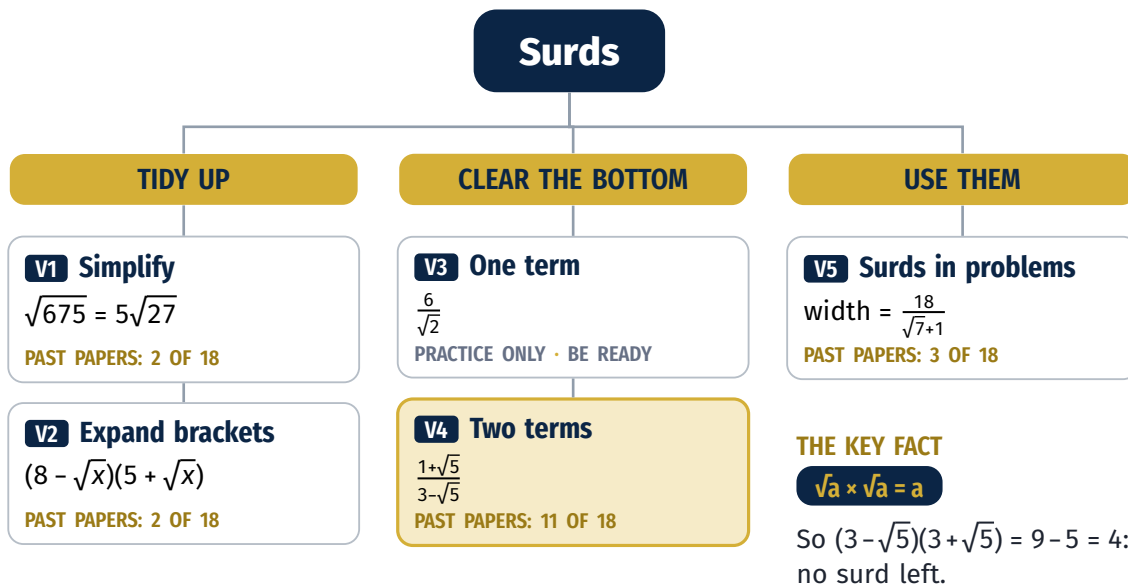
PRACTICE QUESTIONS

YOU WILL MASTER**V1** Simplify **V2** Expand brackets **V3** Rationalise: one term**V4** Rationalise: two terms **V5** Surds in problems

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Surd at a glance



Your notes


★ How often does each variant come up?

16 past-paper questions (Jan 2020 → Nov 2025) hold 18 parts; 27 practice parts — grouped by type.

	PAST PAPERS (OUT OF 18)		PRACTICE PARTS
V1 Simplify	2		5
V2 Expand brackets	2		9
V3 Rationalise: one term	0		4
V4 Rationalise: two terms	11		9
V5 Surds in problems	3		0

WHAT THIS TELLS YOU “Show that $\frac{c}{a-\sqrt{b}}$ can be written as $c + \sqrt{d}$ ” is on almost every Paper 1 (11 of 18 parts, 3 marks). Master the conjugate and V4 is yours.

BEFORE YOU START

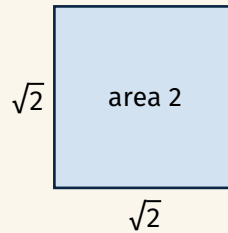
The big idea

**A surd is an exact root.**

$\sqrt{2} = 1.4142 \dots$ never ends, so we keep it as $\sqrt{2}$ — **exact**.

Surds behave like letters:

$3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}$ (like terms), but $\sqrt{2} + \sqrt{3}$ stays as it is.



$\sqrt{2} \times \sqrt{2} = 2$
a square of area 2
has side $\sqrt{2}$

The rules
Multiply

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

$$\sqrt{2} \times \sqrt{8} = 4$$

Divide

$$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$$

$$\frac{\sqrt{12}}{\sqrt{3}} = 2$$

Square

$$\sqrt{a} \times \sqrt{a} = a$$

$$(\sqrt{5})^2 = 5$$

Never

$$\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$$

$$\sqrt{9+16} = 5, \text{ not } 7$$

Square numbers to spot

4

9

16

25

36

49

64

81

100

**CALCULATOR FIRST, THEN CHECK**

Check any surd answer by decimals: $\frac{1+\sqrt{5}}{3-\sqrt{5}} = 4.236 \dots$ and $2 + \sqrt{5} = 4.236 \dots$ ✓. But these are non-calculator questions — the marks are for each stage written out.

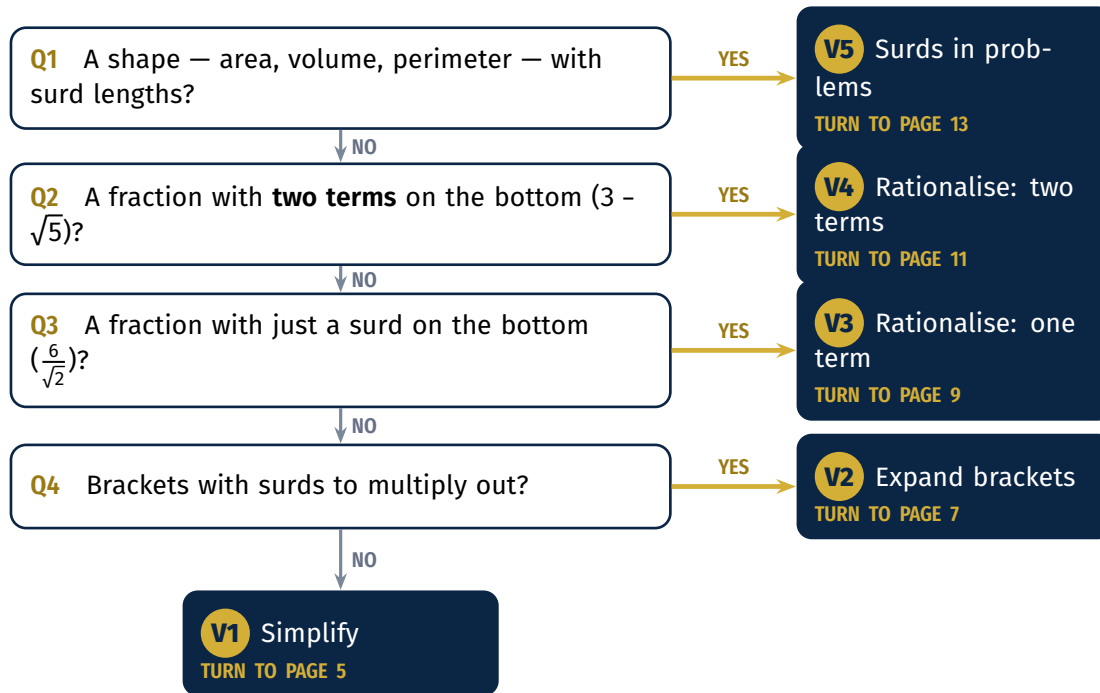
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Simplify a Surd

Split off the biggest square factor; collect like surds.

SEEN IN PAST PAPERS

2 / 18

PRACTICE QUESTIONS: 5

 Your notes

Came up in: Nov 2024 P1H Q17(a) • Jun 2025 P1H Q15(a)

? What is it?

In plain words

Find the **biggest square number** that divides the number: $50 = 25 \times 2$, so $\sqrt{50} = \sqrt{25}\sqrt{2} = 5\sqrt{2}$.

Then add or subtract **like surds**:

$$2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}.$$

IT LOOKS LIKE

$$\sqrt{50}$$

$$\sqrt{12} + \sqrt{27}$$

$$\sqrt{675} = n\sqrt{27}$$

$$\begin{array}{c} \sqrt{50} \\ \downarrow \text{BIGGEST SQUARE} \\ \sqrt{25} \times \sqrt{2} \\ \downarrow \sqrt{25} = 5 \\ 5\sqrt{2} \end{array}$$

How do I do it?

1 Square factor

Biggest square that divides it: 4, 9, 16, 25, 36, ...

2 Split

$$\sqrt{ab} = \sqrt{a}\sqrt{b}.$$

3 Root it

The square comes out: $\sqrt{25} = 5$.

4 Collect

Add like surds (same number under the root).

BIGGEST SQUARE OUT $\sqrt{50} = 5\sqrt{2}$, not $\sqrt{25}\sqrt{2}$.

V1 Simplify a Surd

VARIANT 1 • SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

"Simplify $\sqrt{50}$ " / "write in the form $a\sqrt{b}$ "

→ **Biggest square factor out**

" $\sqrt{675}$ in the form $n\sqrt{27}$ "

→ $675 \div 27 = 25$, **so** $n = 5$

$(\sqrt{3})^5$

→ **Pairs make 3:** $3 \times 3 \times \sqrt{3}$ **WORKED EXAMPLE** **PAST PAPER**

November 2024 • Paper 1H • Q17(a) • 1 mark

Express $\sqrt{675}$ in the form $n\sqrt{27}$ where n is a positive integer.

STEP 1 $675 \div 27 = 25$

◀ the factor next to 27

STEP 2 $\sqrt{675} = \sqrt{25} \times \sqrt{27}$

◀ split

STEP 3 $= 5\sqrt{27}$

◀ root it

ANSWER $n = 5$ $\checkmark 5\sqrt{27} = \sqrt{675} \checkmark$ **WATCH OUT**

$$\times \sqrt{12} + \sqrt{27} = \sqrt{39}$$

$$\checkmark 2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$$

Roots don't add under one sign. Simplify each, then collect.

EXAM TIP

If the number under the root still has a square factor, you are not finished.

Check yourself

calculator first • answers on page 15

1.1 Jun 2025 P1H

$$(\sqrt{3})^5 = k\sqrt{3}. \text{ Find } k.$$

Answer:

1.2 Dr Eslam

$$\text{Simplify } \sqrt{50}.$$

Answer:

1.3 Dr Eslam

$$\text{Simplify } \sqrt{12} + \sqrt{27}.$$

Answer:

V2

VARIANT 2

Expand Brackets with Surds

Grid as in algebra; $\sqrt{a} \times \sqrt{a} = a$; collect whole numbers and surds.

SEEN IN PAST PAPERS

2 / 18

PRACTICE QUESTIONS: 9



Your notes

Came up in: Jan 2021 P1H Q18 • Jan 2022 P2HR Q13

? What is it?

In plain words

Expand exactly like algebra (Topic: Expanding Brackets). Treat $\sqrt{x}$ like a letter — except $\sqrt{x} \times \sqrt{x} = x$.

Then group: **whole numbers** together, **surds** together.
Compare with the form asked.

IT LOOKS LIKE

$$(8 - \sqrt{x})(5 + \sqrt{x})$$

$$(2 + \sqrt{3})^2$$

	5	$+\sqrt{x}$
8	40	$8\sqrt{x}$
$-\sqrt{x}$	$-5\sqrt{x}$	$-x$

$(40 - x) + 3\sqrt{x}$

How do I do it?

1 Grid

Every term times every term.

2 Surd $\times$ surd

$$\sqrt{x} \times \sqrt{x} = x; \quad \sqrt{8} \times \sqrt{8} = 8.$$

3 Collect

Whole numbers together, surds together.

4 Match the form

Compare with $y\sqrt{x} + 21$: numbers with numbers, surds with surds.

$$\sqrt{x} \times \sqrt{x} = x$$

Then it is ordinary expanding.

V2 Expand Brackets with Surds

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

" $(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$, find x and y "

→ **Expand, match numbers and surds**

"Expand and simplify $(2 + \sqrt{3})^2$ "

→ **Square · double · square**

$(a - b)(a + b)$ with surds

→ **DOTS: $a^2 - b^2$** **WORKED EXAMPLE** **PAST PAPER**

January 2021 · Paper 1H · Q18 · 3 marks

$(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$ where x is prime and y is an integer. Find x and y .

STEP 1 $40 + 8\sqrt{x} - 5\sqrt{x} - x$

◀ grid

STEP 3 $(40 - x) + 3\sqrt{x}$

◀ collect

STEP 4 $40 - x = 21 \Rightarrow x = 19, \quad y = 3$

◀ match numbers and surds

ANSWER $x = 19, y = 3$

✓ 19 is prime ✓

WATCH OUT

$$\times (2 + \sqrt{3})^2 = 4 + 3 = 7$$

$$\checkmark 4 + 4\sqrt{3} + 3 = 7 + 4\sqrt{3}$$

A squared bracket has a middle term.

EXAM TIP

Write the four products before collecting — a slip later still earns the method mark.

Check yourself

calculator first · answers on page 15

2.1 Jan 2022 P2HR

$a = \sqrt{8} + 4, b = \sqrt{8} - 4$.
 $(a - b)(a + b)$ can be written as $y\sqrt{4y}$. Find y .

Answer:

2.2 Dr Eslam

Expand and simplify
 $(2 + \sqrt{3})^2$.

Answer:

2.3 Practice · Medium

Expand and simplify
 $(\sqrt{12} - \sqrt{3})^2$.

Answer:

V3

VARIANT 3

Rationalise: One Term

A surd on the bottom: multiply top and bottom by that surd.

SEEN IN PAST PAPERS

0 / 18

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 4


Your notes
? What is it?**In plain words**

“Rationalise the denominator”
means: **no surd on the bottom.**

Multiply top and bottom by the **same surd** — that is multiplying by 1, so the value doesn't change. Then simplify.

$$\frac{6}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{6\sqrt{2}}{2} = 3\sqrt{2}$$

×1: VALUE UNCHANGED

IT LOOKS LIKE

$$\frac{6}{\sqrt{2}} \quad \frac{10}{\sqrt{5}} \quad \frac{1}{27\sqrt{3}}$$

How do I do it?**1 Multiply**

Top and bottom by the surd on the bottom.

2 Bottom

$\sqrt{2} \times \sqrt{2} = 2$ — now a whole number.

3 Top

Multiply out.

4 Simplify

Cancel common factors; simplify any surd.

× THE SURD, TOP AND BOTTOM Multiplying by 1 changes the look, not the value.

V3 Rationalise: One Term

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Rationalise the denominator of $\frac{6}{\sqrt{2}}$ " → $\times \frac{\sqrt{2}}{\sqrt{2}}$

A number in front: $\frac{1}{27\sqrt{3}}$ → **Multiply by $\frac{\sqrt{3}}{\sqrt{3}}$ only**

"Simplify your answer" → **Cancel the number on the bottom**

WORKED EXAMPLE DR ESLAM

Rationalise • 2 marks

Rationalise the denominator of $\frac{6}{\sqrt{2}}$.

STEP 1 $\frac{6}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$

◀ multiply top and bottom

STEP 2 $\frac{6\sqrt{2}}{2}$

◀ bottom: $\sqrt{2} \times \sqrt{2} = 2$

STEP 4 $3\sqrt{2}$

◀ simplify

ANSWER $3\sqrt{2}$

✓ 4.2426... both ways ✓

WATCH OUT

$\times \frac{6}{\sqrt{2}} = \frac{6}{2} = 3$

$\checkmark \frac{6\sqrt{2}}{2} = 3\sqrt{2}$

You must multiply the top too, or the value changes.

EXAM TIP

After rationalising, look for a common factor between the number on top and the bottom.

Check yourself

calculator first • answers on page 15

3.1 Practice • Medium

Rationalise $\frac{10}{\sqrt{5}}$.

Answer:

3.2 Practice • Hard

Rationalise $\frac{1}{27\sqrt{3}}$.

Answer:

3.3 Dr Eslam

Rationalise and simplify $\frac{12}{\sqrt{3}}$.

Answer:

V4**VARIANT 4** MOST COMMON IN THE EXAM

Rationalise: Two Terms

Multiply top and bottom by the conjugate (flip the sign) — DOTS clears the bottom.

SEEN IN PAST PAPERS

11 / 18

PRACTICE QUESTIONS: 9



Your notes

Came up in: Oct/Nov 2020 P1H Q19 • Nov 2021 P2H Q17 • Jan 2022 P2H Q17 • Jun 2022 P1HR Q21 • Jun 2022 P1H Q16 • Jun 2023 P2HR Q16 • May 2024 P1H Q17 • Nov 2024 P1H Q17(b) • Jun 2025 P1H Q15(b) • Jun 2025 P2HR Q16 • Nov 2025 P1H Q17

? What is it?

In plain words

Bottom $3 - \sqrt{5}$: multiply by its **conjugate** $3 + \sqrt{5}$ (same numbers, **opposite sign**).

The bottom becomes $(3 - \sqrt{5})(3 + \sqrt{5}) = 9 - 5 = 4$ — DOTS, the surds cancel.

$$\frac{1 + \sqrt{5}}{3 - \sqrt{5}} \times \frac{3 + \sqrt{5}}{3 + \sqrt{5}}$$

$$\text{bottom: } (3 - \sqrt{5})(3 + \sqrt{5}) = 9 - 5 = 4$$

FLIP THE SIGN — DOTS

IT LOOKS LIKE

$$\frac{1 + \sqrt{5}}{3 - \sqrt{5}}$$

$$\frac{8}{\sqrt{5} - 1}$$

$$\frac{\sqrt{12}}{\sqrt{3} + 2}$$

How do I do it?

1 Conjugate

Bottom $a - \sqrt{b} \rightarrow$ multiply by $a + \sqrt{b}$ (top and bottom).

2 Bottom

DOTS: $a^2 - b$ — a whole number.

3 Top

Expand the brackets (grid).

4 Simplify

Divide every term by the bottom; match the form.

FLIP THE SIGN • DOTS The most common surd question.

V4 Rationalise: Two Terms

VARIANT 4 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Show that $\frac{1+\sqrt{5}}{3-\sqrt{5}}$ can be written as $a + \sqrt{b}$ "

$$\rightarrow \times \frac{3+\sqrt{5}}{3+\sqrt{5}}$$

A square on the bottom: $(\sqrt{2}-1)^2$

$\rightarrow$ **Expand it first: $3 - 2\sqrt{2}$, then conjugate**

WORKED EXAMPLE **PAST PAPER**

May 2024 • Paper 1H • Q17 • 3 marks

Show that $\frac{1+\sqrt{5}}{3-\sqrt{5}}$ can be written in the form $a + \sqrt{b}$ where a and b are integers.

STEP 1 $\frac{(1+\sqrt{5})(3+\sqrt{5})}{(3-\sqrt{5})(3+\sqrt{5})}$ $\leftarrow$ conjugate, top and bottom

STEP 2+3 bottom = $9 - 5 = 4$, top = $8 + 4\sqrt{5}$ $\leftarrow$ DOTS; expand

STEP 4 $\frac{8+4\sqrt{5}}{4} = 2 + \sqrt{5}$ $\leftarrow$ divide every term

ANSWER $2 + \sqrt{5}$ ($a = 2, b = 5$)

✓ both = 4.236... ✓

WATCH OUT

✗ bottom: $(3-\sqrt{5})(3+\sqrt{5}) = 9 + 5$

✓ = $9 - 5 = 4$

DOTS subtracts: $a^2 - (\sqrt{b})^2$.

EXAM TIP

11 of the 18 past-paper parts. Four lines: conjugate, bottom, top, divide — one mark each stage.

Check yourself

calculator first • answers on page 15

4.1 Nov 2021 P2H

Express $\frac{8}{\sqrt{5}-1}$ in the form $\sqrt{a} + b$.

Answer:

4.2 Jun 2023 P2HR

Show that $\frac{2\sqrt{3}}{\sqrt{3}-1}$ can be written as $a + \sqrt{a}$.

Answer:

4.3 Oct/Nov 2020 P1H

Rationalise the denominator of $\frac{6}{3-\sqrt{7}}$ and simplify.

Answer:

V5

VARIANT 5

Surds in Problems

Lengths with surds: area $\div$ length, volume $\div$ area — then rationalise.

SEEN IN PAST PAPERS

3 / 18

PRACTICE QUESTIONS: 0


 Your notes

Came up in: Jan 2020 P2H Q20 • Nov 2021 P1H Q19 • Jun 2023 P2H Q23

? What is it?**In plain words**

Set up the formula (area = length $\times$ width, volume = $x^2 \times$ length ...), make the unknown the subject — it is usually a

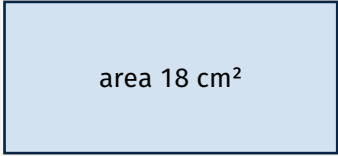
fraction with a surd on the bottom — then **rationalise** (V4).

A square root of the answer may be needed: spot $(a + \sqrt{b})^2$.

IT LOOKS LIKE

$$\text{width} = \frac{18}{\sqrt{7}+1}$$

$$x^2 = 9 + 4\sqrt{5}$$



$$w = \frac{18}{\sqrt{7}+1}$$

How do I do it?**1 Formula**

Area, volume or perimeter in terms of the unknown.

2 Rearrange

Unknown = a fraction.

3 Rationalise

Conjugate, DOTS (V4).

4 Form

Write as $a\sqrt{b} + c$ (or take the square root if it is x^2).

FORMULA $\rightarrow$ FRACTION $\rightarrow$ RATIONALISE Non-calculator: every stage shown.

V5 Surds in Problems

VARIANT 5 · SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO

"The area is 18 cm^2 ; the length is $(\sqrt{7} + 1) \text{ cm}$. Find the width"

→ $18 \div (\sqrt{7} + 1)$, **rationalise**

"Find the value of x ... in the form $a + \sqrt{b}$ " (from x^2)

→ **Spot** $(a + \sqrt{b})^2 = a^2 + b + 2a\sqrt{b}$

"Without using a calculator"

→ **Exact surds all the way** **WORKED EXAMPLE** **PAST PAPER**

January 2020 · Paper 2H · Q20 · 3 marks

A rectangle has area 18 cm^2 and length $(\sqrt{7} + 1) \text{ cm}$. Without a calculator, find the width in the form $a\sqrt{b} + c$.

STEP 1 $w = \frac{18}{\sqrt{7} + 1}$ ◀ area $\div$ length

STEP 3 $\frac{18(\sqrt{7} - 1)}{(\sqrt{7} + 1)(\sqrt{7} - 1)} = \frac{18(\sqrt{7} - 1)}{6}$ ◀ conjugate; bottom $7 - 1 = 6$

STEP 4 $3(\sqrt{7} - 1) = 3\sqrt{7} - 3$ ◀ simplify

ANSWER $3\sqrt{7} - 3 \text{ cm}$

✓ $(3\sqrt{7} - 3)(\sqrt{7} + 1) = 21 - 3 = 18$ ✓

 WATCH OUT

✗ $w = 18 - \sqrt{7} - 1$

✓ $w = \frac{18}{\sqrt{7} + 1}$

Width = area **divided** by length. **EXAM TIP**Check by multiplying back: your width $\times$ the length must give the area exactly. **Check yourself**

calculator first · answers on page 15

5.1 Jun 2023 P2H

A cuboid: square cross-section x by x , length $(2\sqrt{5} - 3)$, volume $(13 + 6\sqrt{5})$. Find x as $a + \sqrt{b}$.

Answer:

5.2 Dr Eslam

A square has area 12 cm^2 . Find its side as a simplified surd.

Answer:

5.3 Dr Eslam

A right-angled triangle has shorter sides $\sqrt{5}$ and $\sqrt{11}$. Find the hypotenuse.

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py).

1.1 $k = 9$

1.2 $5\sqrt{2}$

1.3 $5\sqrt{3}$

2.1 $y = 8$

2.2 $7 + 4\sqrt{3}$

2.3 3

3.1 $2\sqrt{5}$

3.2 $\frac{\sqrt{3}}{81}$

3.3 $4\sqrt{3}$

4.1 $\sqrt{20} + 2$

4.2 $3 + \sqrt{3}$

4.3 $9 + 3\sqrt{7}$

5.1 $x = 2 + \sqrt{5}$

5.2 $2\sqrt{3} \text{ cm}$

5.3 4

THE WHOLE TOPIC ON ONE PAGE

Surds cheat sheet



Your notes
V1 Simplify 2/18

$$\sqrt{50} = 5\sqrt{2}$$

Biggest square out; collect like surds.

V2 Expand 2/18

$$(40 - x) + 3\sqrt{x}$$

Grid; $\sqrt{x}\sqrt{x} = x$; match.**V3 One term**

$$\frac{6}{\sqrt{2}} = 3\sqrt{2}$$

× the surd, top and bottom.

V4 Two terms 11/18

$$\frac{1+\sqrt{5}}{3-\sqrt{5}} = 2 + \sqrt{5}$$

Conjugate: flip the sign, DOTS.

V5 Problems 3/18

$$\frac{18}{\sqrt{7}+1} = 3\sqrt{7} - 3$$

Formula, fraction, rationalise.

**TOP 3 LOST MARKS**

1. $\sqrt{a} + \sqrt{b} = \sqrt{a+b} \times$
2. DOTS bottom $9 + 5 \times$
3. multiplying only the bottom $\times$

BOTTOM LINE $\sqrt{a}\sqrt{a} = a$ · biggest square out · one surd: × it · two terms: conjugate + DOTS.